MODEL QUESTION PAPER
B.Sc(Hons) Physics Major
MG3DSEPHY205 FOUNDATIONS OF THEORETICAL PHYSICS

Duration: 2 Hours **Max. marks : 70**

Part A

(Answer any TEN questions. THREE marks each)

1. Provide an example of a fourth order differential equation of degree 2.
2. Is $xy^2dx + x^2y\ dy = 0$ an exact differential equation?
3. What is the singular solution to a Clairaut's equation $y = px + \frac{1}{p}$?
4. On the minimum of a functional $I = \int_{x_1}^{x_2} F(y, y', x)dx$, when does $\left(F - y'\frac{\partial F}{\partial y'}\right)$ become a constant?
5. What are Lagrange multipliers?
6. Find the Euler-Lagrange equations for a functional $I = \int_{x_1}^{x_2} (y'^2 + z'^2)\, dx$, with constraint $y'z' + kz = 0$.
7. If T_{ij}, U_{ij} are components of order 2 tensors and a_i are components of a vector, does the term $T_{ij}U_{ij}a_i$ represent a tensor, vector or a scalar?
8. Is Kronecker delta the same in coordinate systems related by rotations?
9. If $S_{ij} = S_{ji}$ and $A_{ij} = -A_{ji}$, evaluate $S_{ij}A_{ji}$.
10. Write Dirichlet conditions for the existence of a Fourier series for a function.
11. State Parceval identity for a function that has a Fourier series.
12. State *true* or *false* by giving proper reason : Fourier series of $f(x) = x^2 + x$, where $-\pi \leq x \leq \pi$ will have no sine terms and Fourier coefficients $b_n = 0$ for every n.

Part B

(Answer any 8 questions. FIVE marks each)

13. Solve $\frac{dy}{dx} + \frac{y}{x} = x$, given $y(x = 1) = 0$.
14. Write down wave equation, diffusion equation, Poisson equation and Schrödinger equation.

15. A general solution to Laplace equation has the form

$$\Phi(x,y,z) = \sum_k \sum_l \left(A_{kl} \mathrm{e}^{\sqrt{k^2+l^2}x} + B_{kl} \mathrm{e}^{-\sqrt{k^2+l^2}x} \right) (C_k \sin ky + D_k \cos ky) (E_l \sin lz + F_l \cos lz)$$

Find the solution that satisfies the boundary conditions

$$\Phi(x, y=0, z) = \Phi(x, y=a, z) = \Phi(x, y, z=0) = \Phi(x, y, z=a) = 0 \ ;$$
$$\lim_{x\to\infty} \Phi(x,y,z) = 0 \text{ and } \ \Phi(x=0,y,z) = 5$$

16. Find the differential equation whose solution gives the minimum of the integral $I = \int_{x_1}^{x_2} \frac{1}{2} (y'^2 + y^2) \ dx$, given $\delta y(x_1) = 0 = \delta y(x_2)$.

17. Find the minimum area of the surface of revolution about y axis of a curve between (x_1, y_1) and (x_2, y_2).

18. Components of a second order (rank 2) tensor $\mathbf{T}$ in a Cartesian coordinate system are

$$\mathbf{T} = \begin{bmatrix} 2 & \sqrt{3} & 0 \\ \sqrt{3} & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Find its components in another Cartesian coordinate system by rotating about z-axis through 30°.

19. Given the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$, evaluate $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$

20. Find the Fourier series for

$$f(x) = \begin{cases} 0 & \text{for } 0 \le x \le \pi; \\ 1 & \text{for } \pi \le x \le 2\pi \end{cases}$$

21. For $0 \le x \le 2\pi$, find Fourier series for $f(x) = \sin^3 x - \sin^2 x$ in complex basis.

22. The function $f(x) = |x|$, for $-\pi \le x \le \pi$ has Fourier series

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{m=0}^{\infty} \frac{\cos(2m+1)x}{(2m+1)^2}.$$

$$\text{Evaluate} \quad \sum_{m=0}^{\infty} \frac{1}{(2m+1)^4}.$$